

2D Lattice Defect Detection
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1. Introduction

Whether it be for Semiconductor Manufacturing, Biomedical Device design, or the ever-growing field of Nanotechnology, materials such as Quartz, Silicon, and even Diamond remain relevant for their unique electronic properties. However, as electronic systems continue to downsize, so do the materials they rely on; thus, the need for a reliable characterization tool has become more crucial than ever.

The engineering system developed for this project is an automated computational characterization tool designed to evaluate a series of 2D crystal lattices for structural integrity using the Discrete Fourier Transform. By correctly identifying irregularities in these lattices such as vacancies, line defects, and lattice distortions without the need for costly equipment like an SEM or STM, we hope to be able to prove the DFT's efficacy as a tool for testing the reliability of simulated lattice structures.

There are a plethora of pitfalls associated with designing an algorithm for Lattice characterization- most notably determining the optimal spatial sampling rate required to avoid the phenomenon of aliasing. Aliasing occurs when the sampling frequency is too low to capture the high-frequency variations of the lattice, leading to artifacts that obscure real data.

Physically, atomic lattices exhibit such periodic, high-frequency behavior because atoms naturally arrange themselves in repeating, symmetrical configurations to achieve a state of minimum total potential energy. It is due to this fact that the spatial arrangement of atoms is inherently gaussian, allowing us to take advantage of the discrete, harmonic, high-frequency spikes it creates when transformed into the frequency domain via DFT. As such, analyzing any deviations from these expected spikes should in theory allow the engineer to assess the state of the lattice, as well as identify any irregularities, or "Defects", within the Lattice. A Defect can be as simple as a stretching or compression of the lattice, which would shift the location of the harmonic spikes, or as complex as total entropy within the lattice - a common, naturally-occurring defect known as a Disorder.

2. Data Collection and Experimental Methodology

To ensure that the data captured is relevant to the design problem while minimizing extraneous factors, the experiment utilizes a synthetic lattice generation model. In this model, each atom is represented mathematically as a 2D Gaussian kernel, which accurately simulates the blurred appearance of atoms in high-resolution microscopy.

Unfortunately for us, because we are working with the finite boundaries of an image, such a simulation was rife with unnatural noise caused by spectral leakage from the edges of our lattice image. As such, we opted to apply a 2D Hanning window to the lattice before it undergoes frequency analysis. Not to be confused with the similarly named “Hamming” Window, a Hanning window is a specialized mathematical function that minimizes spectral leakage by completely masking the edges of the Frequency bins created by the DFT. Without this window, the sharp transition at the edge of the image would be interpreted by the Fourier transform as high-frequency noise, creating artificial streaks in the frequency plot. The 1D Hanning window is defined by the expression $w(n) = 0.5(1 - \cos(\frac{2\pi n}{N-1}))$, where the pixel index n and the window length N are used to compute the window weight $w(n)$. We extended this transformation to two dimensions, by implementing it as the product of two orthogonal 1D windows. The weight of the data equals 1 at the center of the window (or grid) and 0 at the corners, ensuring that the periodic features of the lattice are emphasized while the disruptive effects of the image boundaries are suppressed.

3. Mathematical Analysis: The Discrete Fourier Transform

The core of our characterization algorithm involves translating spatial image data into the frequency domain via the Discrete Fourier Transform (DFT). To understand the 2D DFT used in this tool, one must first consider the 1D DFT, which is defined as

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}}$$

Where the strength of different spatial frequencies are compared across each pixel. Once that is done, you can think about the 2D DFT as nothing more than an application of the 1D transform along both dimensions of the image. First, a 1D DFT is performed on every row of the $N \times M$ image matrix. The result of this operation is then transformed again by performing a 1D DFT on every column of the intermediate matrix. Mathematically, this combination results in the 2D DFT equation:

$$F(k, l) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f(n, m)e^{-j2\pi(\frac{kn}{N} + \frac{lm}{M})}$$

In an ideal, defect-free lattice, the periodic spacing between atoms results in high-magnitude spikes known as Bragg Peaks. These peaks appear at specific spatial frequencies defined by the inverse of the lattice spacing.

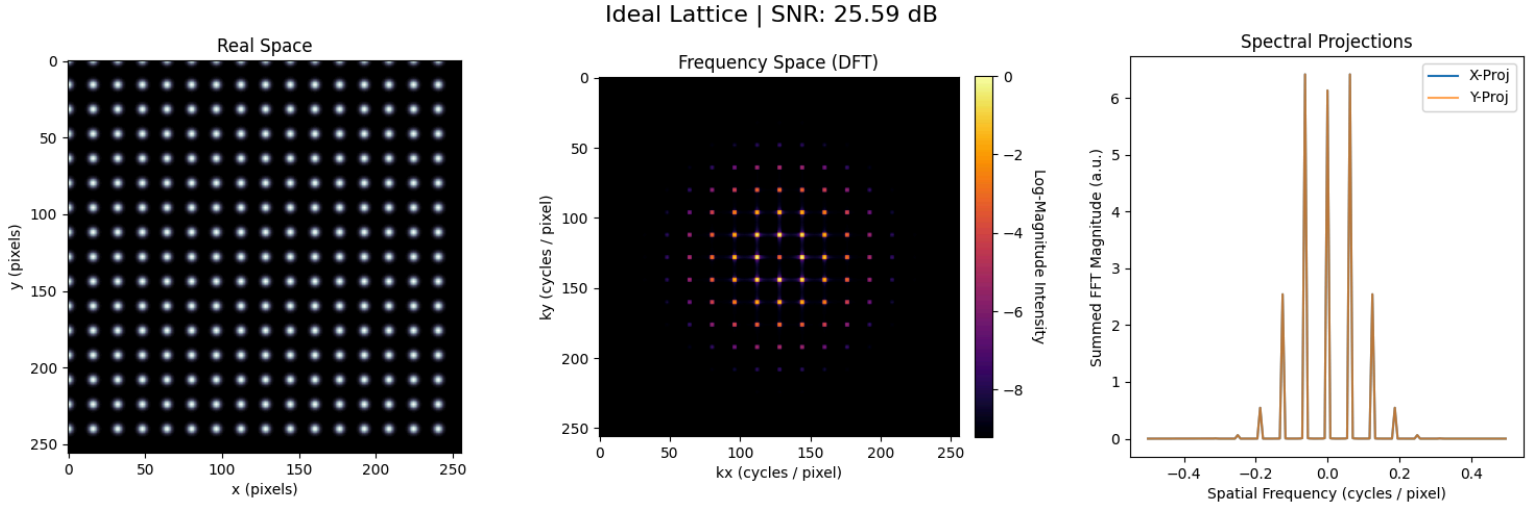


Fig. 1. Our generated grid shows a 256 pixel x 256 pixel perfect lattice. The second subgraph shows the DC component ($f=0$) which is centered at the (128,128) location. All other atoms are displaced 16 pixels apart from this baseline, highlighting how the fundamental frequency f_0 is 0.0625 cycles/pixel. Since our atoms are generated as localized gaussians, they are represented as sharp bumps. Thus, atoms closer to the edge of the lattice are integer multiples of the fundamental frequency, which can be seen in the peaks of the third subgraph. The clear distinction between the spectral projections across both the x and y axes and high SNR are expected of an ideal lattice.

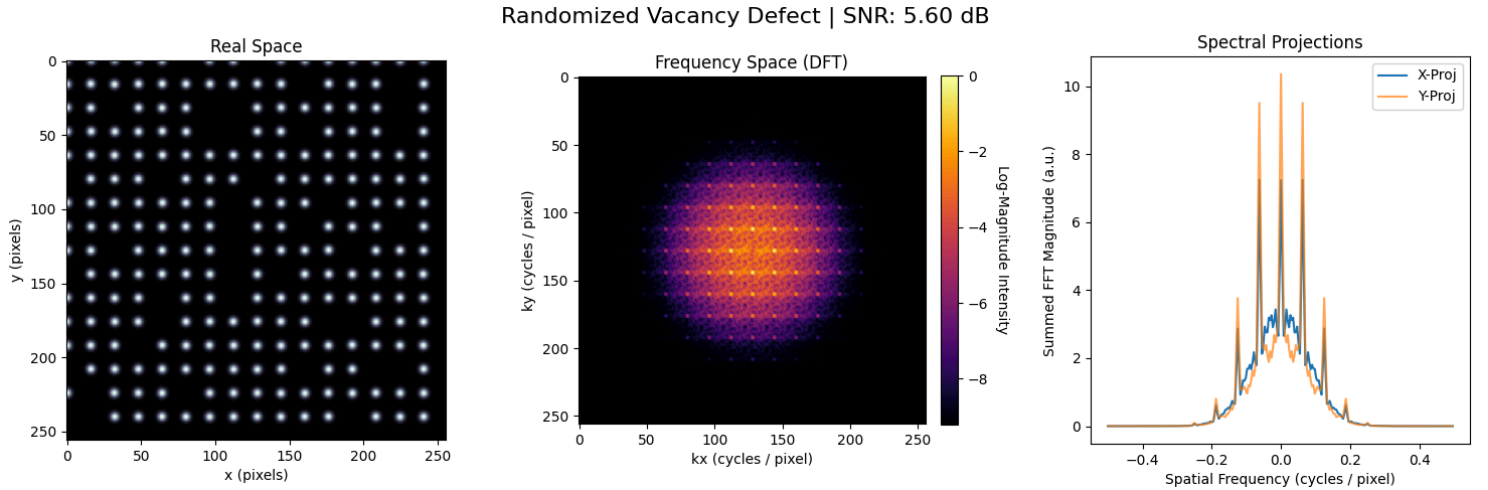


Fig. 2. Although the fundamental frequency and its multiples are depicted in the second subgraph, noise has increased significantly due to the vacancy of atoms causing energy to randomly scatter non-periodically. The SNR has moderately decreased and the floor of the spectral projections graph has risen. Still, lattice symmetry remains.

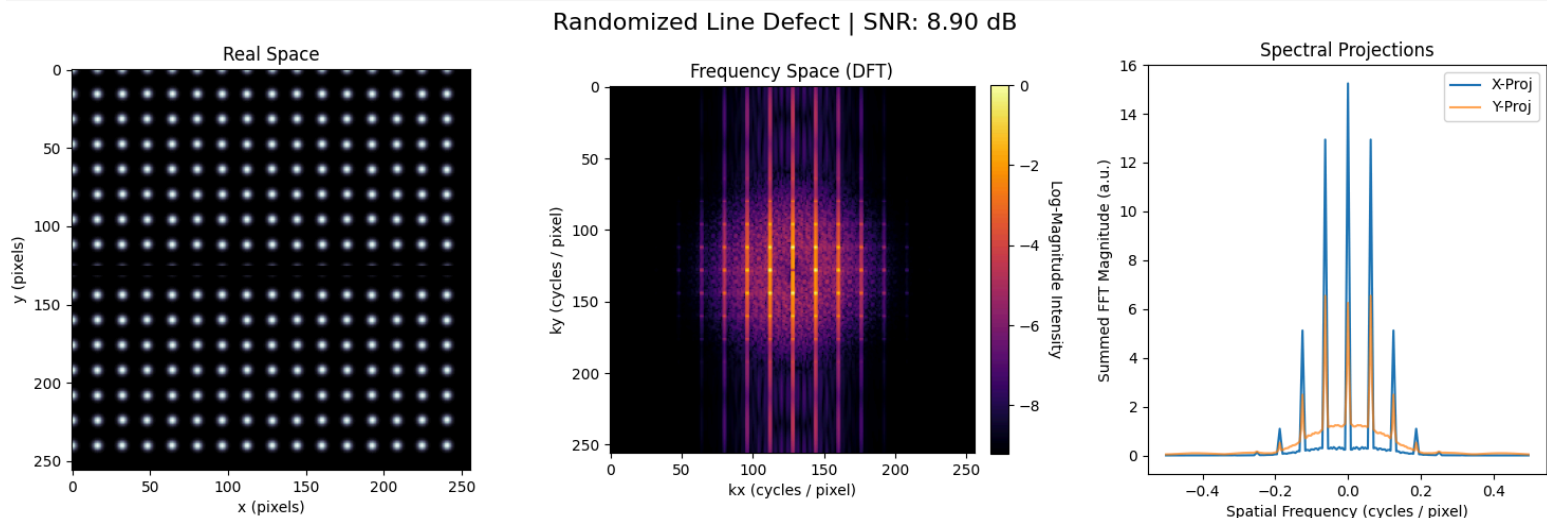


Fig. 3. Line defects create “streaks” in the DFT since symmetry is broken in only the y-axis as well as slight noise. Therefore, the x-axis maintains the same frequencies as the ideal lattice. This defect also moderately reduces SNR.

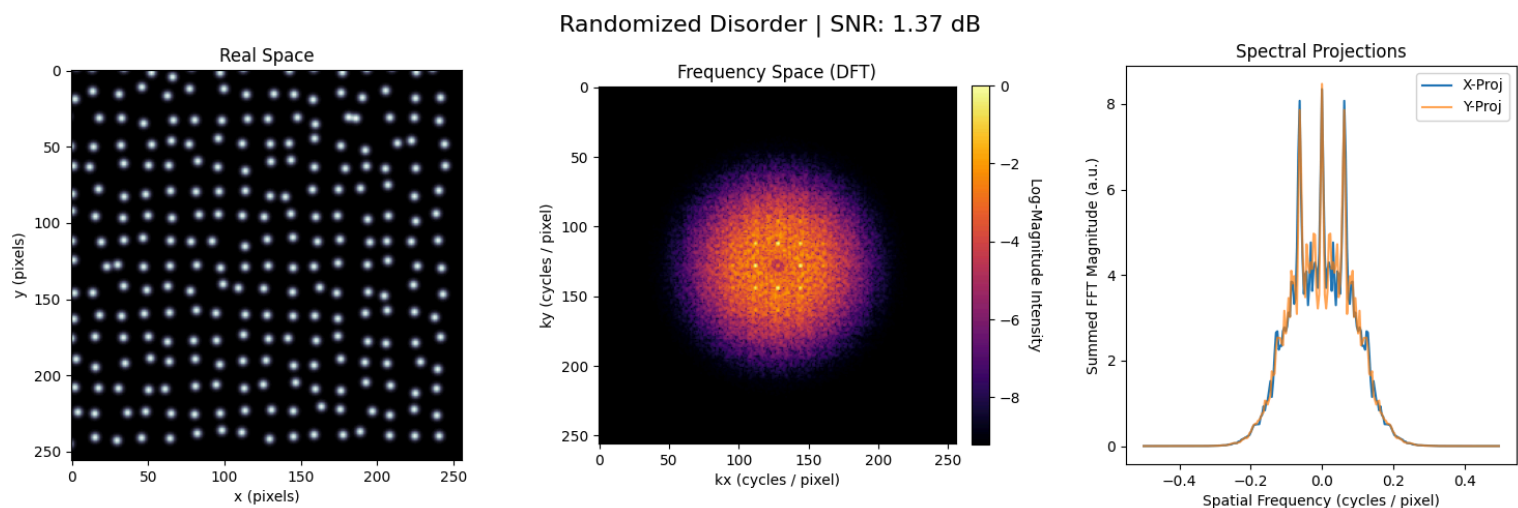


Fig. 4. A disordered lattice analyzed using the DFT shows an extremely noisy result. This is to be expected, as with each atom displaced from its natural location, the SNR ratio is severely compromised, to the point of dampening the Bragg's Peaks.

4. Defect Characterization Metrics and SNR

Our script identifies anomalies by calculating the SNR (Signal-to-Noise Ratio) of the lattice and assessing the distribution of spectral energy. Traditionally, the SNR is used to compare the level

of a desired signal to the level of background noise. In this case, however, the "signal" is the energy contained within the sharp Bragg Peaks, while the "noise" is the diffuse background energy scattered by vacancies and thermal disorder. A high SNR indicates a healthy, periodic lattice, whereas a decreasing SNR suggests significant lattice degradation.

5. Conclusion and Key Engineering Takeaways

The results of this project demonstrate that frequency domain analysis provides a vastly higher sensitivity for detecting structural irregularities than traditional spatial inspection. The most significant finding was the hierarchy of defect detection ease. Distortions, such as global lattice strain, were the most easily detected because they caused immediate and visible shifts in the position of the Bragg Peaks, allowing for effective and easy characterization that could be confirmed with nothing more than the DFT and a set of eyes.

In contrast, vacancies and line defects proved more elusive because they do not alter the peak locations but instead introduce subtle background noise and directional imbalances. These defects required an additional processing step: Specifically, the creation of a Difference Map. We did this by isolating the Bragg Peaks, performing an Inverse DFT to reconstruct an "ideal" version of the lattice, and subtracting that ideal version from the original data, leaving us with a residual image highlighting only the missing or shifted atoms.

All in all, by utilizing the DFT, we are confident that engineers can characterize lattice health with a degree of precision that makes it an especially viable low-cost option for lattice characterization. With an understanding of Signal Processing tools to support its function with a variety of lattices, it's plausible to say that spectral analysis has a secure future as a tool for precision manufacturing and material science.

6. Code

```
import numpy as np
import matplotlib.pyplot as plt

N = 256
LATTICE_SPACING = 16
ATOM_SIGMA = 2.0

def generate_lattice(mode):
    """0: Ideal, 1: Vacancy, 2: Line, 3: Disorder, 4: Combined"""
    x = np.arange(N)
    X, Y = np.meshgrid(x, x)
```

```

image = np.zeros((N, N))

v_prob = 0.15 if mode in [1, 4] else 0.0
line_defect = True if mode in [2, 4] else False
disorder_std = 2.0 if mode in [3, 4] else 0.1
if mode == 0: v_prob, line_defect, disorder_std = 0.0, False, 0.0

for i in range(0, N, LATTICE_SPACING):
    for j in range(0, N, LATTICE_SPACING):
        if np.random.rand() < v_prob: continue

        dx, dy = (0, 0) if mode == 0 else np.random.normal(0,
disorder_std, 2)
        image += np.exp(-((X-(i+dx))**2 + (Y-(j+dy))**2) /
(2*ATOM_SIGMA**2))

    if line_defect:
        mid = (N // LATTICE_SPACING // 2) * LATTICE_SPACING
        image[mid-2:mid+4, :] = 0

return image - np.mean(image)

def analyze_lattice(image, name):
    # FFT Processing
    window = np.outer(np.hanning(N), np.hanning(N))
    F_raw = np.fft.fftshift(np.fft.fft2(image * window))
    mag = np.abs(F_raw) / np.max(np.abs(F_raw))

    # Filter Metrics
    thresh = np.percentile(mag, 99.6)
    peak_mask = mag > thresh

    # SNR Calculation
    snr = 10 * np.log10(
        np.sum(mag[peak_mask]**2) /
        (np.sum(mag[~peak_mask]**2) + 1e-9)
    )

    # Visualization
    fig, axes = plt.subplots(1, 3, figsize=(18, 5))

```

```

fig.suptitle(f"{name} | SNR: {snr:.2f} dB", fontsize=16)

# Real Space
axes[0].imshow(image, cmap='bone')
axes[0].set_title("Real Space")
axes[0].set_xlabel("x (pixels)")
axes[0].set_ylabel("y (pixels)")

# Frequency Space
im_dft = axes[1].imshow(np.log(mag + 1e-4), cmap='inferno')
axes[1].set_title("Frequency Space (DFT)")
axes[1].set_xlabel("kx (cycles / pixel)")
axes[1].set_ylabel("ky (cycles / pixel)")

cbar = fig.colorbar(im_dft, ax=axes[1], fraction=0.046, pad=0.04)
cbar.set_label('Log-Magnitude Intensity', rotation=270, labelpad=15)

# Spectral Projections
freqs = np.fft.fftshift(np.fft.fftfreq(N))
axes[2].plot(freqs, np.sum(mag, axis=0), label='X-Proj')
axes[2].plot(freqs, np.sum(mag, axis=1), label='Y-Proj', alpha=0.7)
axes[2].set_title("Spectral Projections")
axes[2].set_xlabel("Spatial Frequency (cycles / pixel)")
axes[2].set_ylabel("Summed FFT Magnitude (a.u.)")
axes[2].legend()

plt.subplots_adjust(wspace=0.35)
plt.show()

# Run all trials
modes = {0:"Ideal Lattice", 1:"Randomized Vacancy Defect", 2:"Randomized
Line Defect", 3:"Randomized Disorder", 4:"Combined Defects"}
for m_code, m_name in modes.items():
    img = generate_lattice(m_code)
    analyze_lattice(img, m_name)

```